# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-POLAR-EN · v2.2 · 2026-11-02 |
| Package | star-polar |
| Version | 0.1.0 |
| Capability | star.geodesy.polar-stereographic |
| Maturity | Released & Verified (RELEASED) |
| Python | ⩾3.9 |
| Source commit | dc09f61 |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_polar |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_polar-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_polar-0.1.0-py3-none-any.whl | 6d441ad4367db55db161cdc78a4b4e31f12e07c9222cbf5d6df0fc42dab45972 |
| star_polar.py | bb9766d610909103690216cf054a71c10459517903d5bc66a254df2f1347afff |
| star_polar-0.1.0-source.zip | 3ce56be2bc7f26932df91bacc43524823c3c1b6dc53c10dfdf875bb31e20f946 |

# 5. What it does, and its limits

Polar stereographic projection at either pole with any parallel of true scale, forward, inverse and scale factor, and the UPS grid; within 7e-8 m of PROJ and 5e-9 m of PyGeodesy on 600 cases; reproduces Snyder's published example and refuses out-of-range input.

Polar aspect only: no oblique or equatorial stereographic. `ups_forward` does not restrict the latitude to the UPS zones (north of 84 N, south of 80 S): it is the caller's choice of grid. No MGRS letters. PROJ's `+proj=ups` differs from PyGeodesy and from this module by up to 4.4e-5 m while PROJ's `+proj=stere` agrees to 7e-8 m: recorded as a candidate observation about a third-party library, not investigated and not reported.

- `ps_forward`, `ps_inverse`: Snyder's formulas (USGS PP 1395, 21-33 ... 21-41) for either pole, with the parallel of true scale given by the caller (or the pole) and a scale multiplier k0; the isometric latitude is written without cancellation. Reproduce Snyder's numerical example to the printed 0.1 m.
- `ps_scale`: point scale factor, exactly 1 on the parallel of true scale. `ups_forward`, `ups_inverse`: UPS on WGS-84 (k0 = 0.994, 2,000,000 m false easting and northing).
- Measured on 600 cases at both poles (latitudes 20 to 89.999 deg, true scale at 50 to 90 deg): forward within 7e-8 m and inverse within 5e-8 m of PROJ, scale within 6e-11; UPS within 5e-9 m of PyGeodesy (forward and inverse) and 4.4e-5 m of PROJ's `ups`.
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, a pole other than 'N' or 'S', a latitude more than 60 deg beyond the equator, a parallel of true scale in the other hemisphere or within 1 deg of the equator, an ellipsoid or scale outside the stated ranges.

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_polar_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-polar}: Polar stereographic projection and UPS},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_polar/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts